



Coupole

Reference manual

Version 0.1.0 — 8 October 2026

Free toolbox for students of the « Explorer et Comprendre l'Univers » diploma
(Observatoire de Paris).

OHP image bank · image quality · spectra and series · sites and times.

Contents

1	Overview	2
2	Installation	3
2.1	All-inclusive packages	3
2.2	With the computer's Python	3
2.3	pip and pipx	3
2.4	Updates, uninstallation, data	3
3	Getting started	4
4	OHP image bank module	5
4.1	The bank	5
4.2	Catalogue	5
4.3	Processing	5
4.4	Astrometric solution	6
4.5	Fixed headers	7
4.6	Stacks	7
4.7	Anomalies	8
4.8	Sky map	8
4.9	Online record (optional)	9
5	ASTAP (optional)	11
6	Output formats	13
7	Image quality (optional)	14
8	Spectra and series	15
9	Sites and times	16
10	Cosmology	17
11	Computer and performance	19
12	Command line	20
13	Evolution	25
14	Incident reports and privacy	26
15	Writing a module	27
16	Troubleshooting	28
	Sources	29

CHAPTER 1

Overview

Coupole is a free application (GPL-3.0) written in Python 3.10+, running on Windows, macOS and Linux, with a graphical interface (PyQt6) and a complete command line. Every displayed text exists in French and English; the system language is detected and can be forced (*Language* menu, Preferences, or `-lang fr|en`).

Module	Purpose
OHP image bank	Inventory, catalogue, download, astrometric check, fixed headers, conversion, stackable sets, anomalies, sky map.
Image quality	Optional. FWHM, ellipticity, background, noise, gradient, saturation, trails, sampling.
Spectra and series	Reading and plotting 1D data (radio spectra, light curves, tables).
Cosmology	From redshift to distances, ages, volume, distance modulus, scale (Planck 2018 and others).
Sites and times	Site database, OpenStreetMap map, UTC and local time.
My computer	Hardware diagnosis; minimal example module.

Principles: all long work runs off the GUI thread (the interface stays responsive, with progress and cancellation); an interrupted run resumes where it stopped; nothing is deleted without a trace; no data is sent without consent; optional components (ASTAP, SEP, CuPy) may be missing.

CHAPTER 2

Installation

2.1 All-inclusive packages

The packages bundle Python and every library (PyQt6, numpy, astropy, zstandard, lz4, SEP): nothing else to install. They update themselves. ASTAP stays optional and is not included.

System	File	Installation
Windows 10/11 64-bit	Coupole-Setup-x.y.z.exe	Double-click. SmartScreen warning at first launch (unsigned installer): « More info », « Run anyway ». Installed in the user profile, no administrator rights.
macOS Apple Silicon	Coupole-x.y.z-macos-arm64.dmg	Drag Coupole into Applications. First launch: right click, « Open ».
macOS Intel	Coupole-x.y.z-macos-x86_64.dmg	Same.
Linux x86_64 / ARM64	Coupole-x.y.z-linux-*.tar.gz	Unpack, run <code>installer.sh</code> (copied into <code>~/.local/share/coupole</code> , menu entry).

2.2 With the computer's Python

The scripts in the source folder look for Python 3.10 or newer, create a virtual environment, install the dependencies and create launchers. One question only: Enter to confirm (`-oui` / `-Oui` to skip it).

>_ Command

```
sh install.sh # Linux, macOS
install.bat # Windows (double-click; runs install.ps1)
```

Without Python, or without the `venv` module (Debian: `python3-venv` package), the script says so and explains how to get it. Locations: Linux `~/.local/share/coupole/venv`, launchers in `~/.local/bin`; macOS `~/Library/Application Support/Coupole/venv` and `~/Applications/Coupole.app`; Windows `%LOCALAPPDATA%\Programs\CoupolePython`, Desktop and Start menu shortcuts.

2.3 pip and pipx

>_ Command

```
pipx install git+https://github.com/ARP273-ROSE/coupole
pip install . # from the sources; pip install ".[qualite]" adds SEP
python -m coupole # interface; coupole -help: command line
```

Extras: `qualite` (SEP), `validation` (lxml, xisf: XSD validation and second reader), `gpu` (CuPy), `test` (pytest).

2.4 Updates, uninstallation, data

At startup (can be disabled) the packages check the latest release on GitHub and offer to install it: only the code archive (`coupole-app-x.y.z.zip`, about 1 MB) is downloaded and unpacked by the application, the previous version being restored on failure. A pip/pipx installation shows the command to run. Settings, cache and reports live outside the installation folder and survive updates:

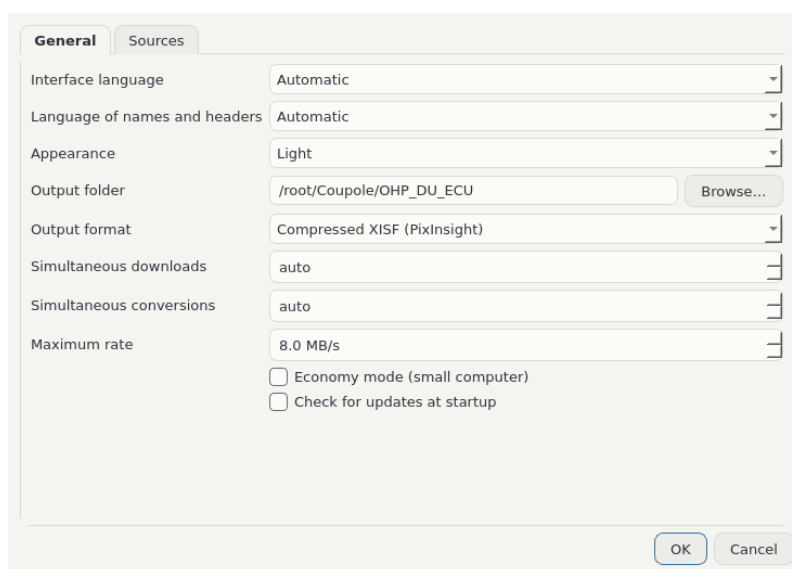
Windows	%LOCALAPPDATA%\Coupole
macOS	~/Library/Application Support/Coupole (cache: ~/Library/Caches/Coupole)
Linux	~/.config/coupole (cache: ~/.cache/coupole)
All	COUPOLE_HOME variable: replaces both (portable installations, tests)

CHAPTER 3

Getting started

The window has a module bar on the left (Ctrl+1 to Ctrl+9) and the current module on the right. Every interactive element has a tooltip. *Help* menu: help on the screen shown (F1), this manual (Shift+F1), keyboard shortcuts, report a problem, incident reports, check for updates, About (version, licence, detected configuration).

F1	help on the screen
Shift+F1	PDF manual
Ctrl+1 ... Ctrl+9	module 1 to 9
Ctrl+,	Preferences
Ctrl+Shift+A	ASTAP assistant
Ctrl+R	OHP bank: refresh the inventory
Ctrl+Q	quit



Preferences (Ctrl+,): interface language; language of folder names, object names and FITS comments; appearance (light or dark theme, Coupole's own and independent of the system theme: fixed style, colours and font size, WCAG contrast ≥ 4.5); output folder; format; parallelism; maximum rate; economy mode; update check. The *Sources* tab lists the service addresses (chapter 13).

CHAPTER 4

OHP image bank module

4.1 The bank

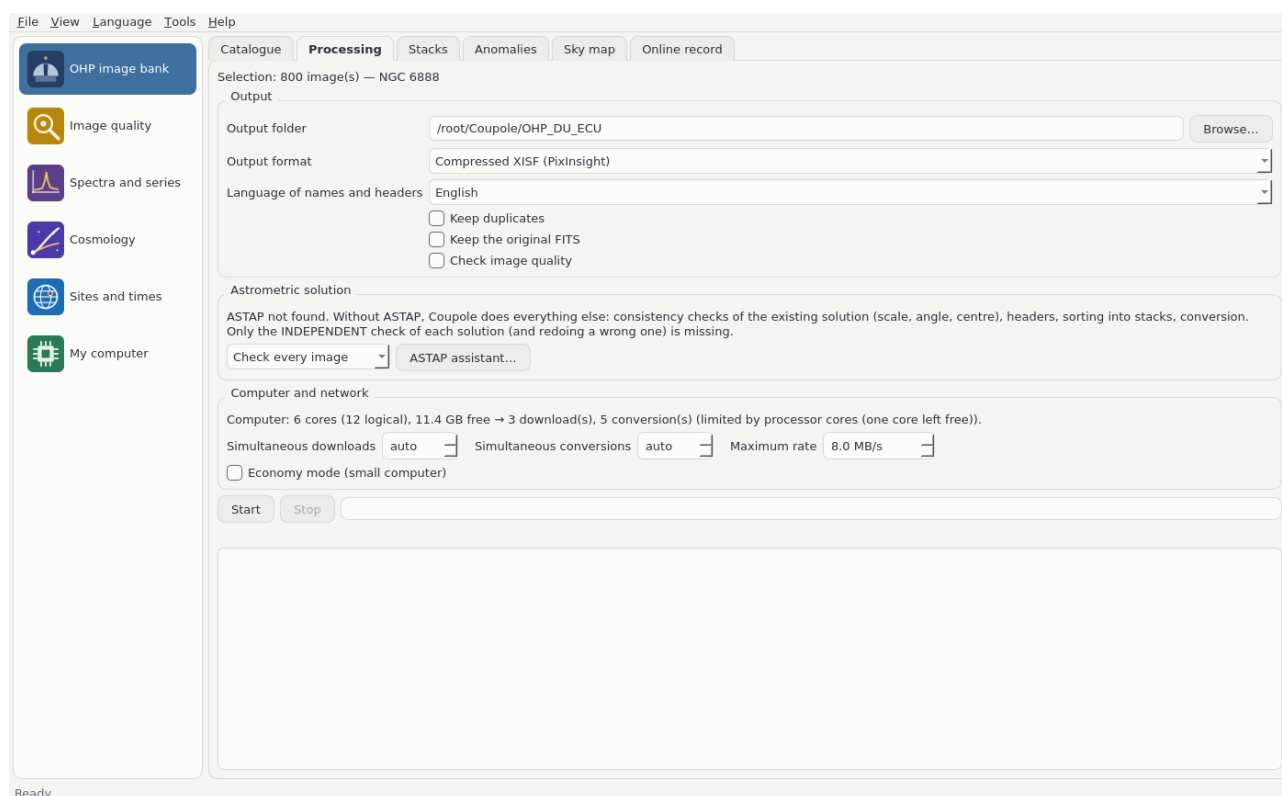
« OHP student observations » gathers the images taken by the diploma's students at the Observatoire de Haute-Provence: 7,989 FITS images on 8 October 2026, 166 objects under 194 names, 60 nights from 2015 to 2026, 78.0 GB; 1.20 m telescope (T120, 7,419 images, 1024 pixels of 0.77", 13.1' field) and the 50 cm IRIS robotic telescope (570 images, 2048 pixels of 0.68" or 4096 pixels of 0.46", 23' and 31' fields). The images are already calibrated (dark, flat) and astrometrically solved: **they must not be recalibrated**.

4.2 Catalogue

The screenshot shows the OHP image bank software interface. The 'Catalogue' tab is active, displaying a table of objects and images. The table has columns for 'type', 'object', 'UTC date', 'site time', 'night', 'tel.', 'filter', and 'exp. (s)'. The table lists various objects, including Planets (Saturn, Uranus), Trans-Neptunian objects and dwarf planets (Pluto, Haumea, Eris, Ritona, Quaoar, 2002 TX300), Planetary nebulae (M27, M57, M76, NGC 6210, NGC 6543, NGC 6826, NGC 7026, NGC 7662), and Nebulae and remnants (Barnard 68, IC 1396, IC 434, IC 5070, IC 5146, M16, M17, M20, M42, NGC 2237, NGC 6888). The table is filtered to show 800 image(s) of 1 object(s) over 1 night(s). The interface also includes a sidebar with navigation icons, a search bar, and a status bar at the bottom.

The inventory comes from the Observatory's public TAP service (<http://tap-ufe.obspm.fr/tap>, table `ivoa.obscore`, one synchronous ADQL query), kept in a cache; a snapshot is shipped for offline work. *Refresh the inventory* (Ctrl+R) queries the service again and flags new images and names (« new since ... »). Database names are grouped into canonical objects; selecting one or several objects lists their images on the right, filtered by night, filter and doubtful dates; this list is the selection to process. The estimate gives the volume to download, the expected output volume and the free space. Times are shown in UTC and in site time (hover: also the user's time).

4.3 Processing



For each image, duplicates excluded:

1. download: at most 4 at once (3 by default), total rate capped (8 MB/s by default), User-Agent naming Coupole, resume by HTTP Range requests, checks of size (server and inventory), multiple of 2,880 bytes and **SIMPLE** card;
2. reading of the FITS cards as written and of the pixels; SHA-1 fingerprint of the pixels (exact duplicates);
3. astrometric solution check (section 4.4);
4. fixed header (section 4.5);
5. writing in the chosen format, reading back and pixel-by-pixel comparison;
6. at the end: sorting into stacks, `LOT.txt`, `INDEX_LOTS.csv`, `_traitement/journal.csv`.

The state is kept in `_traitement/etat.sqlite`: *Stop* then *Start* (or running the same command again) resumes without redoing anything. A folder keeps the name language and format of its first run. *Keep duplicates* also processes what would be left out; *Keep the original FITS* stores them in `_traitement/fits_origine`.

4.4 Astrometric solution

Consistency checks, always made: solution present and readable; scale within 2 % of the nominal one (XPIXSZ/FOCALLEN) or of the night median; angle within 5° of the night median (modulo 180°); centre within 1.5° of the object medoid (fixed) or 2° of the object-and-night medoid (moving). Medians and medoids come from the whole inventory.

With ASTAP (optional), each image is also solved independently:

Status	Meaning
confirmed	ASTAP finds the same centre within 10'' and the same scale within 1 %.
validated	solution kept: checks pass, ASTAP silent (poor or crowded field, short exposure) or absent.
redone	missing or inconsistent solution replaced by ASTAP's (with SIP); the old one goes to HISTORY.
doubtful	checks fail and no ASTAP to decide: image stored apart.
failed	nothing usable: image stored in <code>_sans_solution_astrometrique</code> (<code>_no_astrometric_solution</code> with English names).

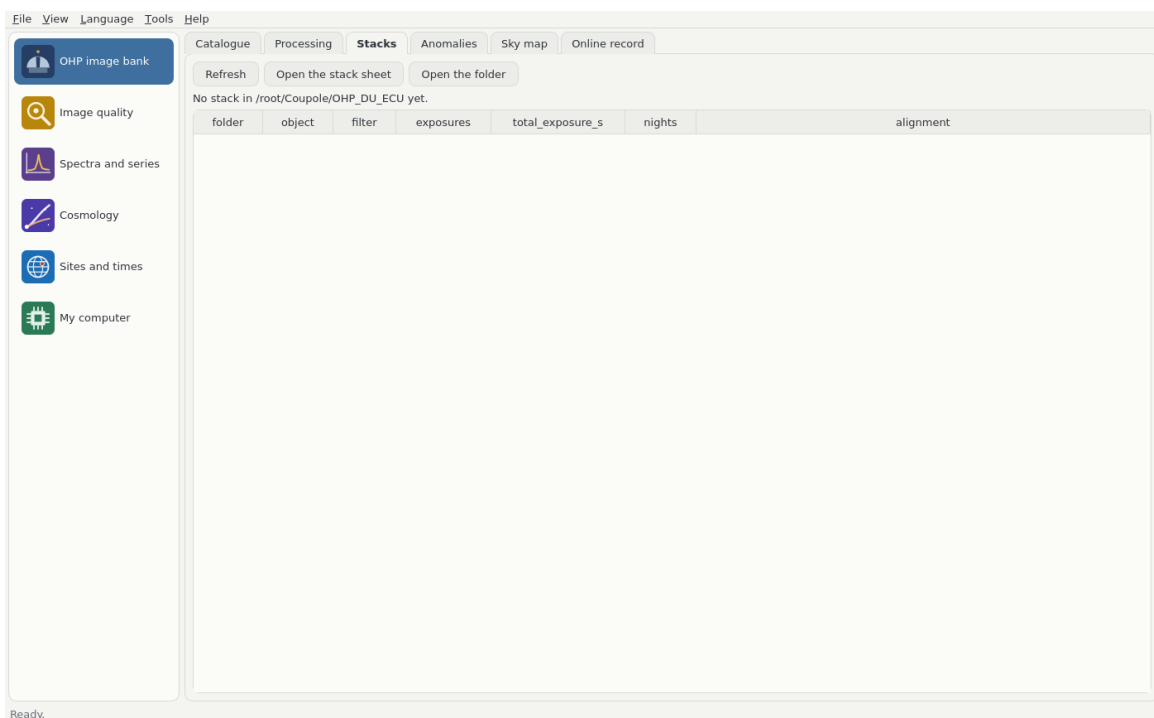
4.5 Fixed headers

All original cards are kept in order; every replaced value is copied into a `HISTORY original value: ... card` (French wording when names are in French).

Keyword	Fix
OBJECT	canonical name (database aliases merged).
FILTER, FILTSYS, FILTERIG	normalised filter (B, V, R, Ha, OIII, g, r, i, z), system (Johnson, Cousins, SDSS, Gunn, Narrowband), original value.
LATITUDE, LONGITUD	fixed only when the swap is proven (swapped values matching the site); correct: untouched; ambiguous: flagged, untouched. Idempotent.
SITELAT, SITELONG, SITEELEV	added when missing (MPC 511 site converted to WGS 84); present and consistent: kept; inconsistent: flagged.
RA, DEC, OBJCTRA, OBJCTDEC	true centre of the solution.
PIXSCALE, ORIENTAT	measured scale, position angle of the Y axis.
FOCALLEN, XPIXSZ	recomputed only when missing or inconsistent by more than 2 %.
CTYPE1/2	« -SIP » without coefficients (IRIS, PinPoint) brought back to TAN.
DATE-OBS, DATEDOUT	added when missing; DATEDOUT = T when the date is shared with another exposure.
XISFCONV	description of the conversion.

The `t_min` column of the database equals `DATE-OBS - exposure/2`; `DATE-OBS` (exposure start) prevails. A header date written in local time (whole-hour offset equal to the site time-zone offset against MJD-OBS/JD, or a « night » exposure dated in daylight) is flagged in the log, never silently fixed.

4.6 Stacks



A stack gathers what is stacked together: same object, same instrument (T120, T120-bin1, T120-sCMOS, IRIS-2048, IRIS-4096), same filter, and same field for a fixed object (all nights) or same night for a moving object.

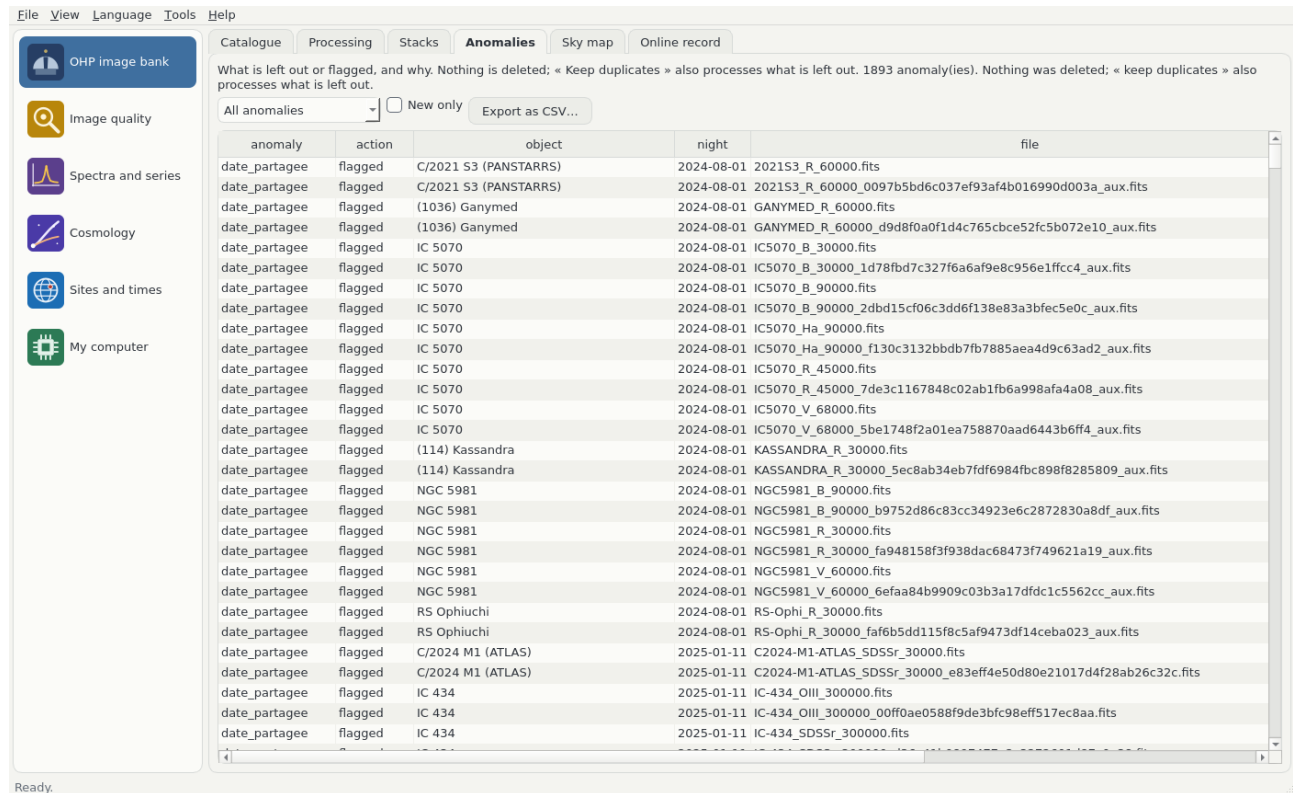
>_ Command

```
07_Nebulae/NGC_6888/field_1_T120/V/20250716-214427_NGC_6888_V_1s.xisf
01_Asteroids/914_Palisana/night_2023-08-15_T120/R/LOT.txt
```

`LOT.txt` (French then English) gives object, database names, instrument, filter, exposures and total exposure, nights, mean centre, angle, solution status, doubtful dates and stacking advice (StarAlignment

or CometAlignment, rejection suited to the number of exposures). File names: UTC date and time of the exposure start, object, filter, exposure; sorting by name = sorting in time. Names are safe under Windows.

4.7 Anomalies



File View Language Tools Help

Catalogue Processing Stacks **Anomalies** Sky map Online record

What is left out or flagged, and why. Nothing is deleted; « Keep duplicates » also processes what is left out. 1893 anomaly(ies). Nothing was deleted; « keep duplicates » also processes what is left out.

All anomalies ☐ New only Export as CSV...

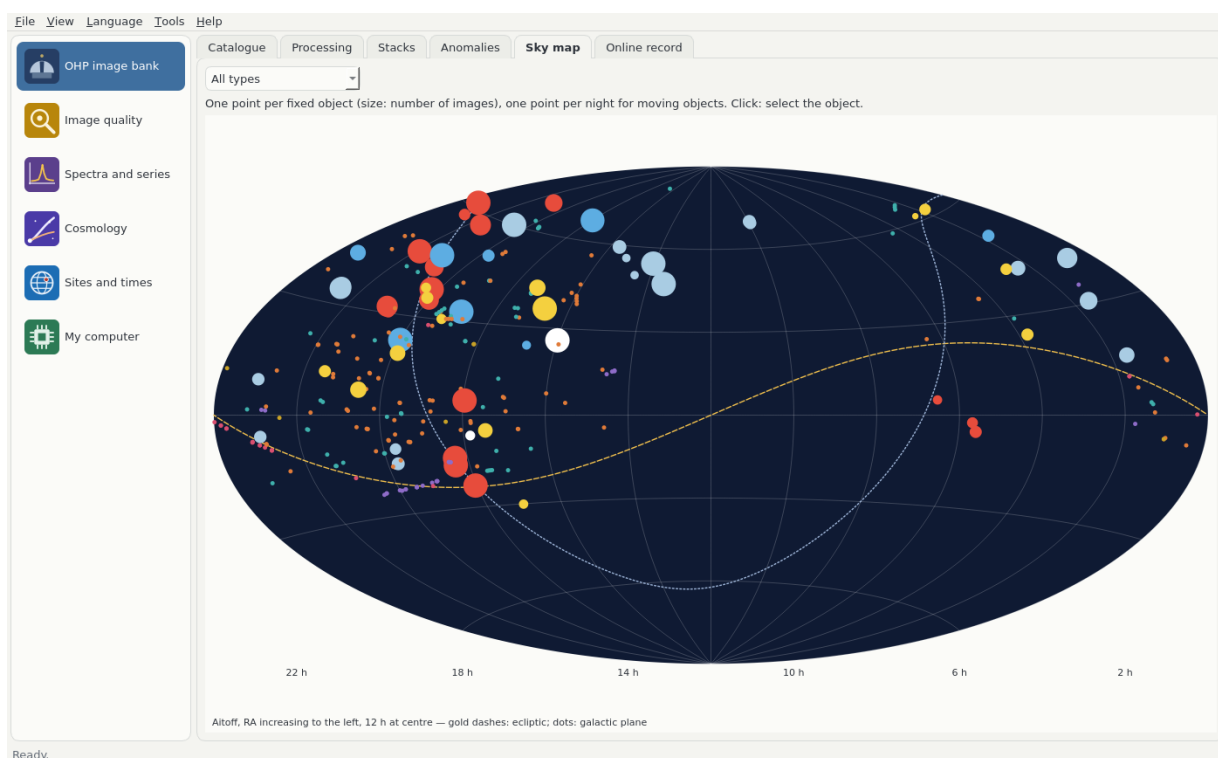
anomaly	action	object	night	file
date_partagee	flagged	C/2021 S3 (PANSTARRS)	2024-08-01	2021S3_R_60000.fits
date_partagee	flagged	C/2021 S3 (PANSTARRS)	2024-08-01	2021S3_R_60000_0097b5bd6c037ef93af4b016990d003a_aux.fits
date_partagee	flagged	(1036) Ganymed	2024-08-01	GANYMED_R_60000.fits
date_partagee	flagged	(1036) Ganymed	2024-08-01	GANYMED_R_60000_d9d8f0a0f1d4c765cbce52fc5b072e10_aux.fits
date_partagee	flagged	IC 5070	2024-08-01	IC5070_B_30000.fits
date_partagee	flagged	IC 5070	2024-08-01	IC5070_B_30000_1d78fbd7c327f6a6a9e8c956e1ffcc4_aux.fits
date_partagee	flagged	IC 5070	2024-08-01	IC5070_B_90000.fits
date_partagee	flagged	IC 5070	2024-08-01	IC5070_B_90000_2dbd15cf06c3dd6f138e83a3bfec5e0c_aux.fits
date_partagee	flagged	IC 5070	2024-08-01	IC5070_Ha_90000.fits
date_partagee	flagged	IC 5070	2024-08-01	IC5070_Ha_90000_f130c3132bbdb7fb7885aea4d9c63ad2_aux.fits
date_partagee	flagged	IC 5070	2024-08-01	IC5070_R_45000.fits
date_partagee	flagged	IC 5070	2024-08-01	IC5070_R_45000_7de3c1167848c02ab1fb6a998afa4a08_aux.fits
date_partagee	flagged	IC 5070	2024-08-01	IC5070_V_68000.fits
date_partagee	flagged	IC 5070	2024-08-01	IC5070_V_68000_5be1748f2a01ea758870aad6443b6ff4_aux.fits
date_partagee	flagged	(114) Kassandra	2024-08-01	KASSANDRA_R_30000.fits
date_partagee	flagged	(114) Kassandra	2024-08-01	KASSANDRA_R_30000_5ec8ab34eb7fdf6984fbc898f8285809_aux.fits
date_partagee	flagged	NGC 5981	2024-08-01	NGC5981_B_90000.fits
date_partagee	flagged	NGC 5981	2024-08-01	NGC5981_B_90000_b9752d86c83cc34923e6c2872830a8df_aux.fits
date_partagee	flagged	NGC 5981	2024-08-01	NGC5981_R_30000.fits
date_partagee	flagged	NGC 5981	2024-08-01	NGC5981_R_30000_fa948158f3f938dac68473f749621a19_aux.fits
date_partagee	flagged	NGC 5981	2024-08-01	NGC5981_V_60000.fits
date_partagee	flagged	NGC 5981	2024-08-01	NGC5981_V_60000_6efaa84b9909c03b3a17dfdc1c5562cc_aux.fits
date_partagee	flagged	RS Ophiuchi	2024-08-01	RS-Ophi_R_30000.fits
date_partagee	flagged	RS Ophiuchi	2024-08-01	RS-Ophi_R_30000_faf6b5dd115f8c5af9473df14ceba023_aux.fits
date_partagee	flagged	C/2024 M1 (ATLAS)	2025-01-11	C2024-M1-ATLAS_SD5Sr_30000.fits
date_partagee	flagged	C/2024 M1 (ATLAS)	2025-01-11	C2024-M1-ATLAS_SD5Sr_30000_e83eff4e50d80e21017d4f28ab26c32c.fits
date_partagee	flagged	IC 434	2025-01-11	IC-434_OIII_300000.fits
date_partagee	flagged	IC 434	2025-01-11	IC-434_OIII_300000_00ff0ae0588f9de3bfc98eff517ec8aa.fits
date_partagee	flagged	IC 434	2025-01-11	IC-434_SD5Sr_300000.fits

Ready.

Kind	Action	Meaning
meme_fichier	left out	same exposure under another path or spelling
copie_diurne	left out	daytime-dated copy of a correctly dated image
pixels_identiques	left out	same pixels (SHA-1) as another image, at processing
memes_meta	flagged	same date, filter, exposure, size, different name
date_partagee	flagged	DATE-OBS identical to another exposure
date_diurne	flagged	Sun above the horizon at the site (never for the Sun or radio data)
champ_incoherent	flagged	centre far from the other exposures of the object
classement	flagged	name classified automatically or not at all
instrument	flagged	instrument missing from the table

On 8 October 2026: $133 + 11 = 144$ duplicates left out, 1,654 shared dates, 3 daytime exposures, 92 inconsistent fields. CSV export with bilingual columns; nothing is deleted.

4.8 Sky map



Aitoff projection, right ascension increasing to the left, 12 h at the centre; one point per fixed object (size: number of images) and per night for moving objects; ecliptic dashed, galactic plane dotted.

4.9 Online record (optional)

HD 192163
Identifier chosen by SIMBAD (CDS) HD 192163
*The requested name was « NGC 6888 »: check that this is the same object (SIMBAD may attach a nebula to its central star).
 Wolf-Rayet (WolfRayet*)*
Type α = 303.02726°, δ = 38.35494° — 20 12 06.54, +38 21 17.7
Coordinates (ICRS, J2000) U = 7.12 B = 7.49 V = 7.50 G = 7.20 J = 6.13 H = 5.90 K = 5.56
Magnitudes 0.5772 ± 0.016 mas → distance = 1733 pc (5651 ly)
Parallax 1732 pc (−48 / +48) — method: paral — 2020yCat.1350...0G
Measured distance v = −100 km/s, z = −0.000334 (quality D)
Radial velocity / redshift WN6(h)-s
Spectral type NAME Crescent Nebula, AG+38 1977, ALS 10934, BD+37 3821, CEL 4971, CSV 102983, DO 18703, EXO 201017+3812.3, GC 28056,
Other names GCRV 12567, GEN# +1.00192163, GSC 03151-01765, HBHA 3703-58, Hen 3-1849 ... and 31 more
Links [SIMBAD page](#) · [Aladin Lite \(sky image\)](#)
 Source: SIMBAD (CDS), retrieved on 2026-10-08 14:56 UTC (cache).

When the Internet is available, the *Online record* tab (or the button of the same name under the catalogue) shows public information on the selected object:

- deep sky and stars: **SIMBAD** (CDS, Strasbourg) — type, ICRS coordinates, magnitudes, parallax (and the distance it gives when precise enough), preferred measured distance with its method and reference, radial velocity and redshift, angular size, spectral and morphological types,

other names; links to the SIMBAD page, **Aladin Lite** (sky image centred on the object) and, for galaxies, **NED**; when SIMBAD does not know the name, **Sesame** (CDS) also queries NED and Vizier;

- asteroids, comets, trans-Neptunian objects: **JPL** Small-Body Database — orbit class, near-Earth or not, elements (a, e, i, q, Q, Ω , ω , period, epoch), diameter, albedo, rotation period, absolute magnitude;
- planets and NEOCP designations: no record (explanatory message).

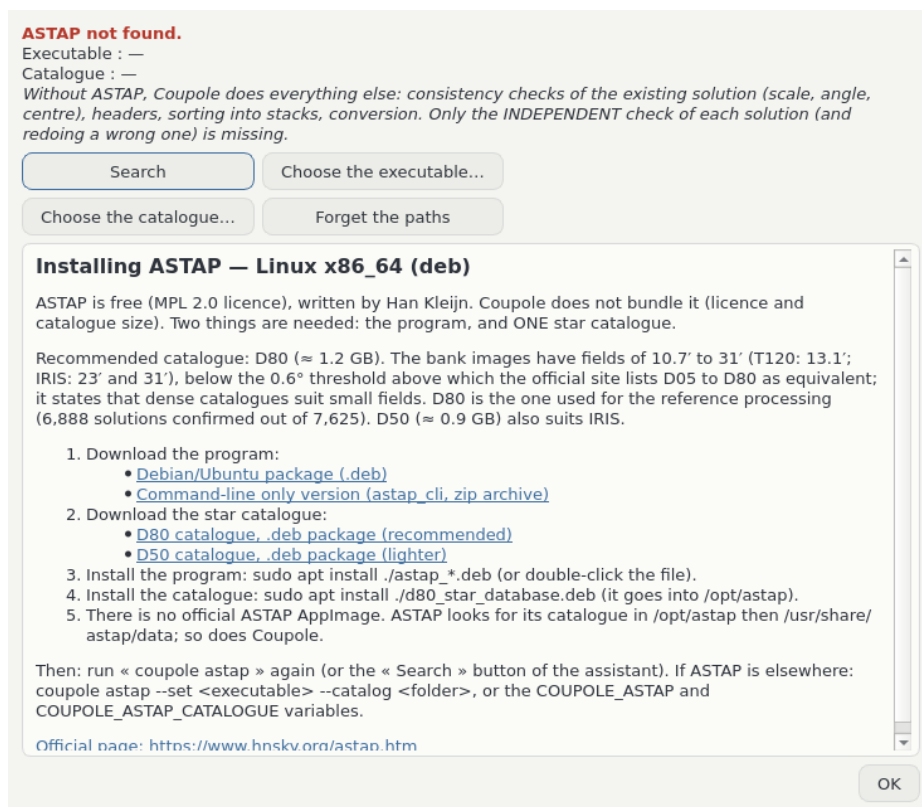
The identifier chosen by the service is always shown: SIMBAD, for instance, attaches « NGC 6888 » to its Wolf-Rayet star HD 192163; the record says so. A positive redshift can be sent to the Cosmology module (*Send this redshift to the Cosmology module* button).

Politeness and offline use. Nothing is sent while the tab is hidden; one request per selected object (after 0.6 s without another selection change), at most one per second and per service, 8 s timeout, Coupole's User-Agent. Answers are kept for 30 days in Coupole's cache (`enligne.json`, with their date; « not found » for one day). Offline, the last known record is shown with its date, otherwise a clear message; the rest of the application never depends on it. The *Query online services* box stops all requests. The addresses (`simbad.base`, `simbad.page`, `aladin.page`, `ned.page`, `sbdb.page`, `resolution.sbdb`, `resolution.sesame`) are set like the other sources.

CHAPTER 5

ASTAP (optional)

Without ASTAP, Coupole does everything else. ASTAP (Han Kleijn, MPL 2.0 licence, <https://www.hnsky.org/astap.htm>) adds an independent check of each solution. It needs the program and one star catalogue.



Catalogue. The official site states that for a field of 0.6° or more, D05, D20, D50 and D80 all suit, and that dense catalogues serve small fields. The bank's fields range from $10.7'$ (0.18°) to $31.4'$ (0.52°): **D80** is recommended (about 1.2 GB; catalogue of the reference processing, 6,888 solutions confirmed out of 7,625); D50 (about 0.9 GB) also suits IRIS.

System	Program (links of the official page)
Windows x64	astap_setup.exe then d80_star_database.exe (same folder, C:\Program Files\astap); optional: astap_command-line_version_win64.zip
Windows ARM64	astap_command-line_version_win11_aarch64.zip and d80_star_database.zip, in one folder
macOS Apple Silicon	astap_M1.pkg, d80_star_database.pkg; once: codesign -force -s - /Applications/ASTAP.app/Contents/MacOS/astap
macOS Intel	astap.pkg, d80_star_database.pkg
Linux Debian/Ubuntu	astap_amd64.deb (or astap_aarch64.deb), d80_star_database.deb (into /opt/astap)
Linux Fedora/openSUSE	astap_amd64.rpm, catalogue d80_star_database.zip unzipped into /opt/astap
Linux Arch	astap_amd64.pkg.tar.zst, catalogue as zip
Other Linux	astap_amd64.tar.gz or astap_command-line_version_Linux_amd64.zip; no official AppImage

Detection. Order: chosen path (Preferences, `coupole astap -set`), COUPOLE_ASTAP variable, PATH (astap_cli, then astap), default locations (Windows Program Files\astap, %LOCALAPPDATA%\astap; macOS /Applications/ASTAP.app, /usr/local/opt/astap; Linux /opt/astap, /usr/bin, /usr/local/bin, ~/astap). Catalogue: chosen folder, COUPOLE_ASTAP_CATALOGUE, program folder,

`/opt/astap`, `/usr/share/astap/data`, `/usr/local/opt/astap`. A catalogue is declared complete with its 1,476 tiles; the version is read from `astap_cli -h`.

CHAPTER 6

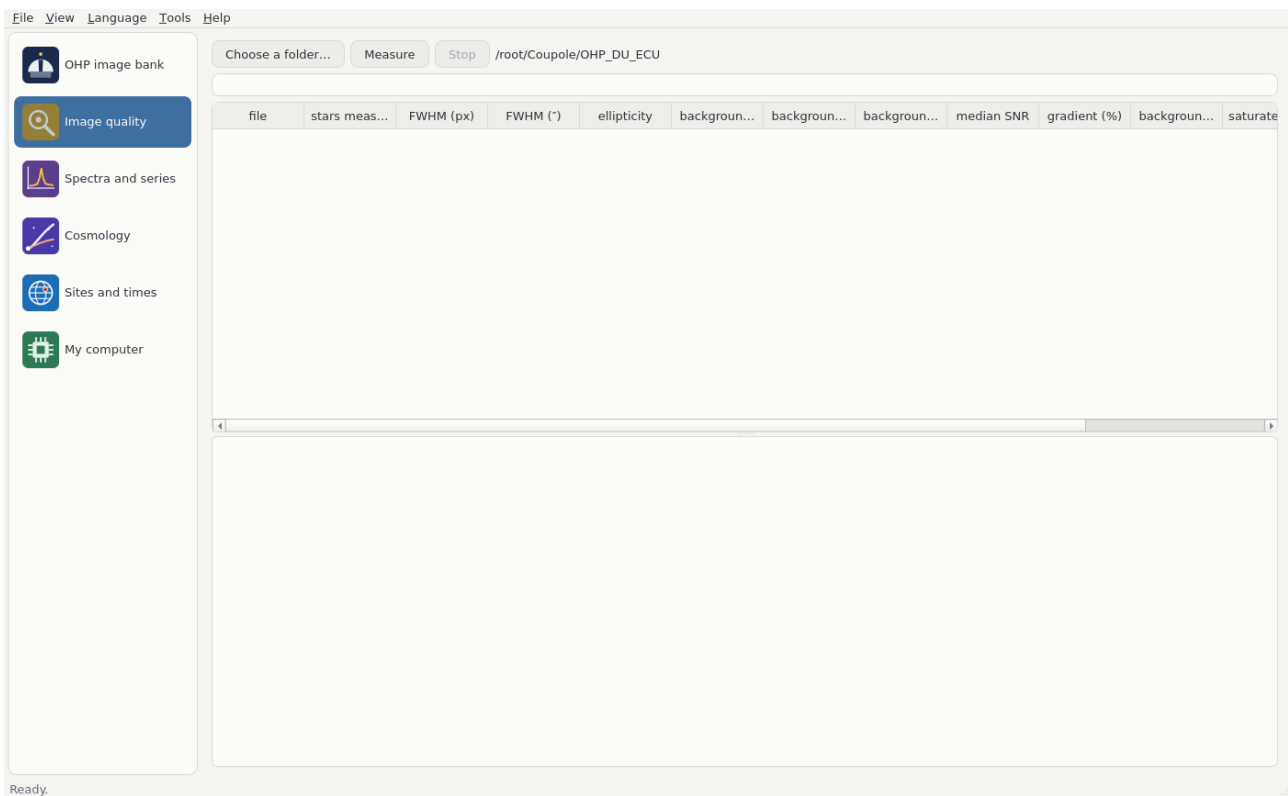
Output formats

Format	Content	Readers
XISF (default)	Float32 in ADU, <code>bounds="-1000:65535"</code> ; UInt16 for integer IRIS FITS; zstd+sh level 9 compression; <code>Observation:*</code> , <code>Instrument:*</code> properties	PixInsight
FITS <code>.fits.fz</code>	lossless tile compression: GZIP_2 without quantization (floats), RICE_1 (integers)	Siril, astropy, DS9
FITS	uncompressed float32 (or UInt16)	all

XISF: `bounds` sets the range PixInsight maps to $[0,1]$ when opening: nothing to set, same scale for every exposure; the $-1,000$ lower bound keeps the negative values of short calibrated exposures. Size: 36 % of the original FITS for the T120 (float64), 62 % for IRIS. **Compressed FITS:** measured with astropy 8.0, RICE_1 and HCOMPRESS_1 quantize floats (one-ADU error, even with `quantize_level=0`); only GZIP_1 and GZIP_2 without quantization restore floats bit for bit. **float32 FITS:** in PixInsight, set the FITS input range to 0–65,535 with *Truncate*. For every format the difference to the original FITS stays below half a float32 ULP (checked pixel by pixel).

CHAPTER 7

Image quality (optional)



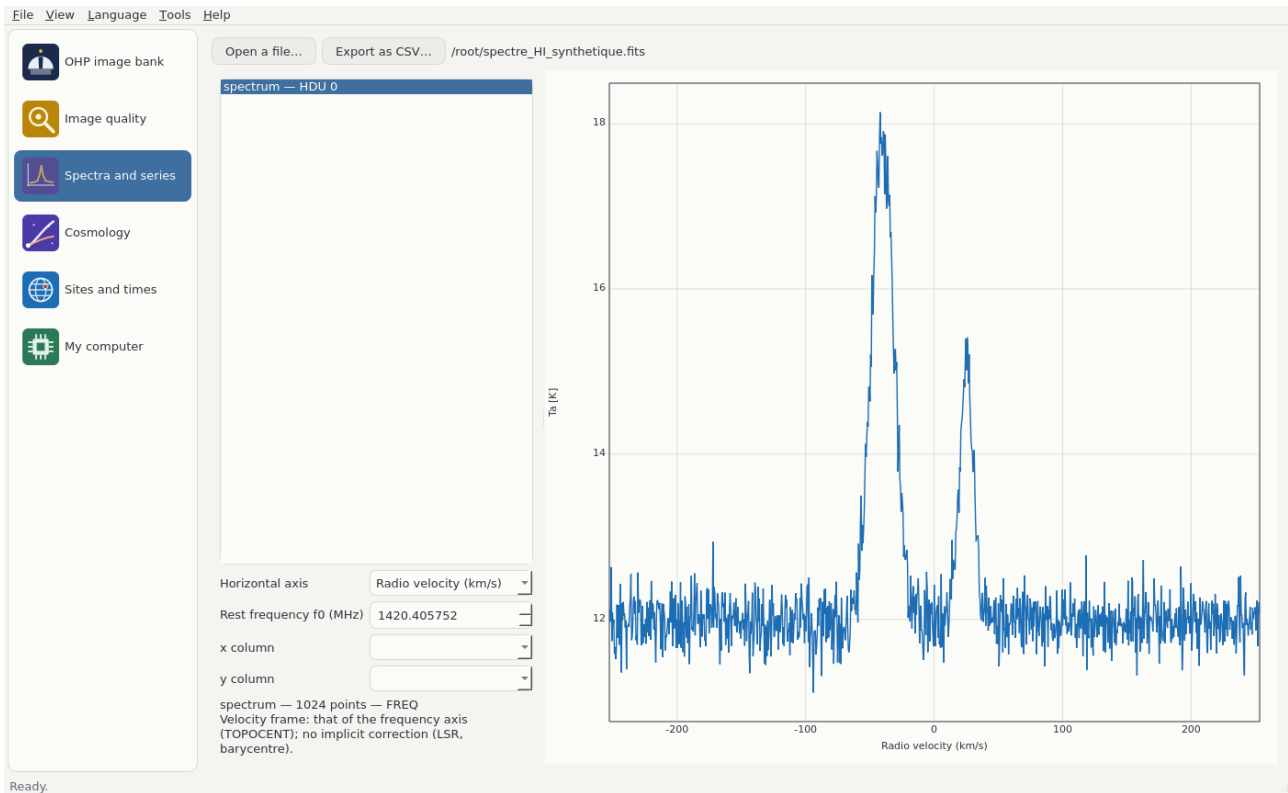
The module needs the SEP library (Source Extractor in Python, LGPL, about 1 MB); it never runs automatically (*Measure* button, *Check image quality* box of the processing, or `coupole qualite FOLDER -write`). Report per stack: `QUALITE.csv` and `QUALITE.txt`.

Measure	Method	Synthetic validation
FWHM	elliptical Moffat with free β fitted to each isolated, unsaturated star (model integrated over the pixel); median	$\leq 0.4 \%$ (Gauss and Moffat, 2.5–8 px)
ellipticity	$1 - b/a$ of the same fit	± 0.02
3×3 map	medians per ninth of the field	order recovered
background, noise	SEP, 64 px mesh	0.5 %; 5 %
gradient, residual	plane fitted to the background map; residual amplitude	gradient and vignetting recovered
saturated	peak above SATURATE/DATAMAX (else 60,000 ADU)	5 out of 5
trails	objects with major axis ≥ 12 px and $a/b \geq 6$	1 out of 1, 0 false
SNR	only with the gain (EGAIN/GAIN)	given or absent
sampling	$\text{FWHM} < 2$ px, 2–5 px, > 5 px	—

On 14 IRIS images (12 nights, FWHM from 2.4 to 11.3 px), the measured FWHM and the one written by PinPoint in the header correlate ($r = 0.974$), median ratio 1.046 (0.85 to 1.16): the two estimators differ (PinPoint’s method is not described). Not offered for lack of validation: mag/arcsec² (no zero point), dust-donut detection. A FWHM growing towards an edge may come from the sensor, field curvature, collimation, but also from seeing, tracking or focus: the map alone cannot tell.

CHAPTER 8

Spectra and series



Formats read: FITS (1D image with a linear CRVAL/CDELTA/CRPIX WCS axis, $1 \times 1 \times N$ radio cubes, binary tables) and CSV or text in columns (optional header, unit in parentheses or brackets). Radio velocity: $v = c(1 - f/f_0)$ (FITS radio convention), f_0 read from RESTFRQ, otherwise the HI line at 1,420.405751768 MHz. The frame is that of the frequency axis (SPECSYS); no LSR or barycentric correction is made implicitly. Formats of teaching radio antennas are added with a reader (CONTRIBUTING.md, « Adding a format »).

CHAPTER 9

Sites and times

The screenshot shows the Coupole application interface. On the left is a sidebar with icons for 'OHP image bank', 'Image quality', 'Spectra and series', 'Cosmology', 'Sites and times' (selected), and 'My computer'. The main window has a menu bar (File, View, Language, Tools, Help) and a toolbar with 'Add a site...', 'Delete', and 'Online map' (unchecked). A 'Show these images' button is in the top right. Below the toolbar is a table of sites:

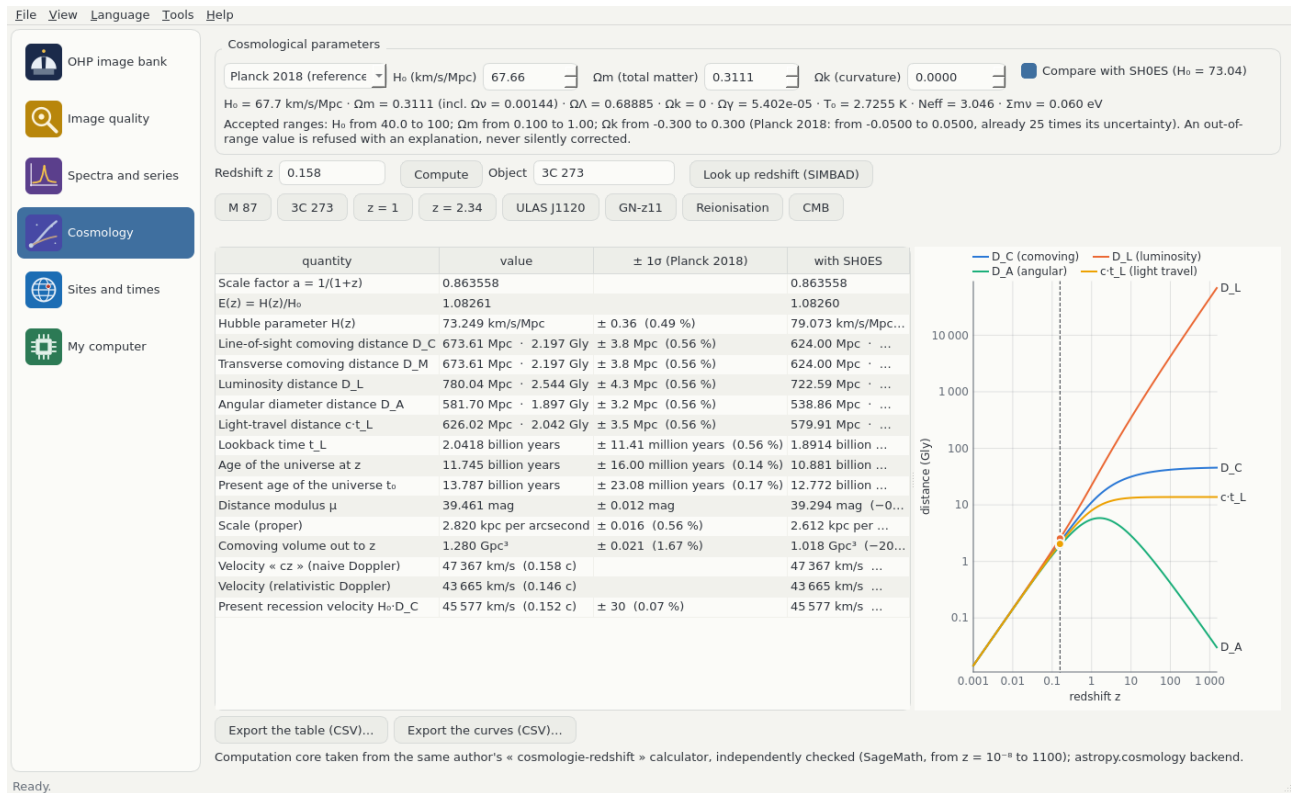
	id	name	latitude (°)	longitude E (°)	altitude (m)	MPC code	time zone	local time	origin
1	ohp	Observatoire de Haute-Provence	43.93187	5.71570	634	511	Europe/Paris	2026-10-08 16:56	shipped
2	meudon	Observatoire de Paris, site de Meudon	48.80507	2.23100	158	005	Europe/Paris	2026-10-08 16:56	shipped
3	paris	Observatoire de Paris	48.83638	2.33675	60	007	Europe/Paris	2026-10-08 16:56	shipped

Below the table is a map showing the locations of the three sites: 'Observatoire de Paris' (red dot), 'Observatoire de Paris, site de Meudon' (red dot), and 'Observatoire de Haute-Provence' (red dot). The map has a light blue background and is labeled 'Offline: plain background (map © OpenStreetMap contributors when the network is back)'. The status bar at the bottom left says 'Ready.'.

Time is always kept in UTC (aware dates); the site's local time and the user's are shown as well. The « evening date » of a night is the site's local date at the previous noon; daylight is computed from the Sun's altitude at the site (astropy, no IERS download). Shipped sites: OHP (MPC 511), Meudon (005), Paris (007), geodetic coordinates computed from the Minor Planet Center parallax constants. Partner sites are added (IANA time zone, MPC code if any) once their coordinates are known from a reliable source. Map: OpenStreetMap tiles cached on disk, at most two downloads at a time, User-Agent naming the application, attribution shown; offline, plain background and points.

CHAPTER 10

Cosmology



The redshift z of a distant object does not give « one » distance but several, all correct, answering different questions. The module computes them in the Λ CDM model with

$$E(z)^2 = \Omega_r(z)(1+z)^4 + \Omega_m(1+z)^3 + \Omega_k(1+z)^2 + \Omega_\Lambda, \quad D_C = \frac{c}{H_0} \int_0^z \frac{dz'}{E(z')},$$

radiation included (CMB photons and neutrinos, one of them massive, 0.06 eV).

Quantity	Use
D_C line-of-sight comoving	present distance of the object; saturates at ≈ 46 Gly (particle horizon)
D_M transverse comoving	equals D_C in a flat universe; enters D_L and D_A
$D_L = (1+z)D_M$	photometry: $F = L/(4\pi D_L^2)$; distance modulus $\mu = 5 \log_{10}(D_L/10 \text{ pc})$
$D_A = D_M/(1+z)$	sizes: $\theta = \ell/D_A$; scale in kpc per arcsecond; maximum near $z \approx 1.6$
$c t_L$	path travelled by the light (the most intuitive, the least physical)
t_L , age at z , t_0	lookback time; check $t_L + t(z) = t_0$
V_C	comoving volume out to z (Hogg 1999, exact in curved universes too)
velocities	cz (naive), relativistic Doppler (inappropriate), $H_0 D_C$ (the only correct one)

Parameters. Planck 2018 by default ($H_0 = 67.66$, $\Omega_m = 0.3111$ incl. $\Omega_\nu = 0.00144$, $T_0 = 2.7255$ K) with the 1σ uncertainty propagated from *correlated* H_0 and Ω_m ($\rho = -0.976$: ignoring it would overestimate the uncertainty) and, optionally, the comparison with SH0ES ($H_0 = 73.04$, Hubble tension). For comparison: Planck 2015, WMAP 9-year, a « textbook » Λ CDM (70 / 0.3 / 0.7 without radiation) and a custom set (H_0 , Ω_m , Ω_k , Planck 2018 radiation). Ranges: H_0 from 40 to 100, Ω_m from 0.1 to 1, Ω_k from -0.3 to 0.3 (± 0.05 for Planck 2018); an out-of-range value is *refused with an explanation*, never silently corrected.

Input. A z (comma or point), one of the examples (M 87, 3C 273, $z = 1$, $z = 2.34$, ULAS J1120, GN-z11, reionisation, CMB), or an object name: its redshift is requested from SIMBAD (tolerant naming: « m87 », « ngc4486 »; several candidates: you choose). $z \leq 0$ (approaching object, like M 31) gives no distance; below $z = 0.03$ the object's peculiar velocity dominates (warning); beyond $z \approx 1090$ the universe is opaque. Table and curves export to CSV (« ; », UTF-8; raw values in Mpc, Gyr, Gpc³).

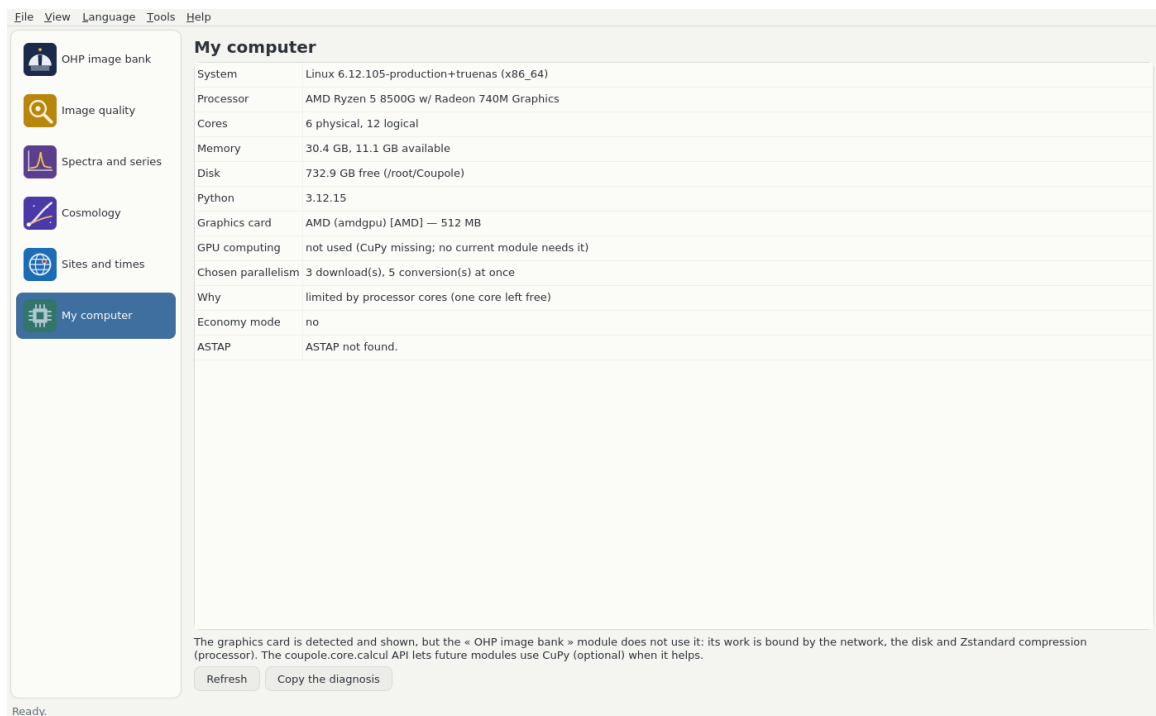
Origin and checks. The computation core is taken from the same author's « cosmologie-redshift » calculator (public repository <https://github.com/ARP273-ROSE/cosmologie-redshift>), with astropy.cosmology. Before integration it was compared on 8 October 2026 with astropy and with an independent SageMath integration (mpmath, 18 digits, exact Fermi-Dirac function for the neutrinos), from $z = 10^{-8}$ to 1100, for $\Omega_k = 0, \pm 0.01, \pm 0.05$: relative deviation $\leq 3 \times 10^{-6}$ on distances and $\leq 2 \times 10^{-5}$ on ages and $E(z)$ (Komatsu fit of the neutrinos in astropy); no bug. astropy's comoving volume, however, loses all its digits at very small z in curved universes (down to a negative volume): Coupole uses a series expansion there, matched to the closed form. Scripts: `outils/verif_cosmo/`.

>_ Command

```
coupole cosmo 0.158 -shoes
coupole cosmo -object "3C 273"
coupole cosmo 1 2 3 -model wmap9 -csv results.csv
coupole cosmo -curves 0.001 1500 300 -csv curves.csv
```

CHAPTER 11

Computer and performance



Detection without mandatory dependency: system, architecture, processor, physical and logical cores (bounded by affinity and container quota), total and available memory, free space, graphics cards (nvidia-smi, `/sys/class/drm`, `system_profiler`, CIM). Parallelism: 3 downloads (at most 4), conversions = physical cores – 1 bounded by available memory at 500 MB per conversion; economy mode (1 conversion, 2 downloads) automatic below 4 GB or 2 cores; manual limits. The graphics card is not used by the OHP bank (work bound by network, disk and Zstandard); `coupole.core.calcul` offers CuPy to future modules.

Measurement (Ryzen 5 8500G, NVMe)	Value
<code>coupole -version</code>	0.1–0.2 s
window shown	0.4–0.5 s
inventory loaded (background thread)	3.1 s at first launch, 0.7 s afterwards
memory peak of a T120 / IRIS 4096 ² conversion	144 MB / 600 MB
interface during 3 downloads (28 MB/s)	max gap of a 20 ms timer: 20.8 ms

CHAPTER 12

Command line

Everything the interface does is available on the command line; `-json` for scripts. Without argument, `coupole` opens the interface. Commands have a French name and an English alias.

>_ Command

```
coupole -lang en ohp estimate "NGC 6888" -night 2025-07-16
coupole -lang en ohp process "NGC 6888" -night 2025-07-16 -filter B -filter V -filter R
-dest ~/OHP -yes
coupole -lang en ohp download Pluto -no-doubtful-dates -dest ~/FITS
coupole -lang en qualite ~/OHP -write
```

coupole

<code>-version</code>	show the version and exit
<code>-lang, -langue {auto,fr,en}</code>	message language (auto: system language)
<code>-v, -verbeux, -verbose</code>	detailed log

coupole astap

Look for ASTAP and its star catalogue, show what to install for this system, or record their location. Coupole works without ASTAP.

<code>-guide</code>	show the installation guide for this system
<code>-definir, -set PATH</code>	path of the ASTAP executable (or its folder)
<code>-catalogue, -catalog FOLDER</code>	star catalogue folder (D80, D50...)
<code>-oublier, -forget</code>	forget recorded paths (back to detection)
<code>-json</code>	JSON output (for scripts)

coupole rapports (reports)

Nothing is sent without consent. Without consent, reports stay in a local folder.

<code>-activer, -enable</code>	allow sending (anonymous) reports
<code>-desactiver, -disable</code>	forbid sending reports

coupole maj (update)

Query the releases published on GitHub. The standalone package updates itself; a pip/pipx installation shows the command to run.

<code>-appliquer, -apply</code>	download and install the update (standalone package)
---------------------------------	--

coupole sources

Without option: list the addresses and their origin (shipped default, remote file, forced value).

<code>-forcer, -set KEY VALUE</code>	force a value (never overwritten by an update)
<code>-default, -default KEY</code>	go back to the default value for this key
<code>-tout-reinitialiser, -reset-all</code>	go back to the default values for every key
<code>-tester, -test KEY</code>	test the connection to this address
<code>-distant, -remote</code>	fetch the sources file published in the repository now

coupole ohp inventaire (inventory)

Bank inventory from the public TAP service (<http://tap-ufe.obspm.fr/tap>, table `ivoa.obscore`), kept in a local cache.

<code>-rafraichir, -refresh</code>	query the Observatory's TAP service again (one request)
------------------------------------	---

coupole ohp catalogue (catalog)

The 194 database names grouped into objects, sorted by type.

<code>-type TYPE</code>	category: ast, neocp, com, pla, tno, pn, neb, amas, gal, eto (repeatable)
<code>-telescope {T120,IRIS}</code>	telescope (repeatable)
<code>-chercher, -search TEXT</code>	keep objects whose name contains this text
<code>-json</code>	JSON output (for scripts)

coupole ohp images (list)

One line per image: date, telescope, filter, exposure, flags (duplicate, shared date, daytime).

<code>OBJECT</code>	object(s): « NGC 6888 », « Pluto », « 1998 AG6 », or a database name
<code>-type TYPE</code>	category: ast, neocp, com, pla, tno, pn, neb, amas, gal, eto (repeatable)
<code>-telescope {T120,IRIS}</code>	telescope (repeatable)
<code>-filtre, -filter F</code>	filter: B, V, R (Johnson and Cousins), Bc, Vc, Rc, Ha, OIII, g, r, i, z, or the exact name (« Johnson R »); repeatable or comma-separated
<code>-nuit, -night YYYY-MM-DD</code>	night (evening date); repeatable
<code>-sans-dates-douteuses,</code>	leave out exposures with a shared DATE-OBS or dated in daytime
<code>-no-doubtful-dates</code>	
<code>-tout, -all</code>	the whole bank (≈ 78 GB)
<code>-csv FILE</code>	also write the list to this CSV file

coupole ohp estimer (estimate)

Estimate before downloading, with the free space at the destination.

<code>OBJECT</code>	object(s): « NGC 6888 », « Pluto », « 1998 AG6 », or a database name
<code>-type TYPE</code>	category: ast, neocp, com, pla, tno, pn, neb, amas, gal, eto (repeatable)
<code>-telescope {T120,IRIS}</code>	telescope (repeatable)
<code>-filtre, -filter F</code>	filter: B, V, R (Johnson and Cousins), Bc, Vc, Rc, Ha, OIII, g, r, i, z, or the exact name (« Johnson R »); repeatable or comma-separated
<code>-nuit, -night YYYY-MM-DD</code>	night (evening date); repeatable
<code>-sans-dates-douteuses,</code>	leave out exposures with a shared DATE-OBS or dated in daytime
<code>-no-doubtful-dates</code>	
<code>-tout, -all</code>	the whole bank (≈ 78 GB)
<code>-format {xisf,fz,fits}</code>	output format: xisf (PixInsight, default), fz (lossless compressed FITS), fits (float32)
<code>-dest FOLDER</code>	destination folder (default: ~/Coupole/OHP_DU_ECU or the one in the settings)

coupole ohp traiter (process)

Full processing of a selection, resumable: running the same command again resumes where it stopped.

OBJECT	object(s): « NGC 6888 », « Pluto », « 1998 AG6 », or a database name
-type TYPE	category: ast, neocp, com, pla, tno, pn, neb, amas, gal, eto (repeatable)
-telescope {T120,IRIS}	telescope (repeatable)
-filtre, -filter F	filter: B, V, R (Johnson and Cousins), Bc, Vc, Rc, Ha, OIII, g, r, i, z, or the exact name (« Johnson R »); repeatable or comma-separated
-nuit, -night YYYY-MM-DD	night (evening date); repeatable
-sans-dates-douteuses,	leave out exposures with a shared DATE-OBS or dated in daytime
-no-doubtful-dates	
-tout, -all	the whole bank ($\approx$ 78 GB)
-dest FOLDER	destination folder (default: ~/Coupole/OHP_DU_ECU or the one in the settings)
-format {xisf,fz,fits}	output format: xisf (PixInsight, default), fz (lossless compressed FITS), fits (float32)
-noms, -names {fr,en}	language of folder names, object names and headers (default: interface language)
-astap {auto,tous,suspectes,jamais,all,suspicious,never}	ASTAP check: auto/all (every image when ASTAP is present), suspicious, never
-telechargements, -downloads N	simultaneous downloads (automatic by default; at most 4)
-conversions N	simultaneous conversions (default: from cores and memory)
-economie, -eco	economy mode: 1 conversion, 2 downloads
-debit, -rate MB/s	maximum download rate (default 8 MB/s)
-garder-fits, -keep-fits	also keep the original FITS (in _traitement/fits_origine)
-garder-doublons, -keep-duplicates	also process duplicates (inventory and identical pixels) instead of leaving them out
-oui, -yes	accept a selection over 5 GB without confirmation

coupole ohp telecharger (download)

Layout: <folder>/<object>/<night>_<telescope>/<original file>.

OBJECT	object(s): « NGC 6888 », « Pluto », « 1998 AG6 », or a database name
-type TYPE	category: ast, neocp, com, pla, tno, pn, neb, amas, gal, eto (repeatable)
-telescope {T120,IRIS}	telescope (repeatable)
-filtre, -filter F	filter: B, V, R (Johnson and Cousins), Bc, Vc, Rc, Ha, OIII, g, r, i, z, or the exact name (« Johnson R »); repeatable or comma-separated
-nuit, -night YYYY-MM-DD	night (evening date); repeatable
-sans-dates-douteuses,	leave out exposures with a shared DATE-OBS or dated in daytime
-no-doubtful-dates	
-tout, -all	the whole bank ($\approx$ 78 GB)
-dest FOLDER	destination folder (default: ~/Coupole/OHP_DU_ECU or the one in the settings)
-telechargements, -downloads N	simultaneous downloads (automatic by default; at most 4)
-debit, -rate MB/s	maximum download rate (default 8 MB/s)
-oui, -yes	accept a selection over 5 GB without confirmation

coupole ohp anomalies (anomalies-report)

Duplicates (same file, daytime copy, identical pixels), shared or daytime dates, inconsistent field, classification to check, unknown instrument. Nothing is deleted.

-dest FOLDER	add the pixel duplicates found in this processing folder
-csv FILE	also write the list to this CSV file
-nouveaux, -new	only images new since the last refresh

coupole ohp classer (classify)

Without argument: list the names to check. With a database name and `-object`: record the correction (merge into an existing object if `-type` is omitted). `-online`: query JPL SBDB and CDS Sesame.

DB_NAME	target name as it appears in the database
<code>-objet, -object OBJECT</code>	canonical object to associate (existing one: merge)
<code>-type {ast,neocp,com,pla,tno,pn,neb,amas,gal,eto,autre}</code>	category: ast, neocp, com, pla, tno, pn, neb, amas, gal, eto (repeatable)
<code>-oublier, -forget</code>	forget the correction of this name
<code>-en-ligne, -online</code>	resolve unknown names online (JPL SBDB, CDS Sesame), with cache

coupole ohp ranger (sort)

Useful after an interruption, or to merge several runs in the same folder.

<code>-dest FOLDER</code>	destination folder (default: <code>~/Coupole/OHP_DU_ECU</code> or the one in the settings)
---------------------------	--

coupole ohp bilan (status)

progress of a processing folder

<code>-dest FOLDER</code>	destination folder (default: <code>~/Coupole/OHP_DU_ECU</code> or the one in the settings)
<code>-json</code>	JSON output (for scripts)

coupole qualite

Optional: background, noise, FWHM and ellipticity (3×3 map), background gradient, saturation, trails, sampling

FOLDER	folder of stacks (or one image file)
<code>-ecrire, -write</code>	write QUALITE.csv and QUALITE.txt in each stack

coupole donnees lire (read)

describe the contents of a file (FITS, CSV)

FILE	file to read
<code>-json</code>	JSON output (for scripts)

coupole donnees exporter (export)

export a spectrum or a series as CSV

FILE	file to read
<code>-csv FILE</code>	export a spectrum or a series as CSV
<code>-hdu HDU</code>	index of the dataset in the file (default: the first 1D one)
<code>-vitesse, -velocity</code>	add the radio velocity (frequency spectrum)
<code>-f0 MHz</code>	rest frequency in MHz (default: RESTFRQ, else H I)

coupole cosmo

From redshift to distances: comoving, luminosity, angular, lookback time, age, volume, distance modulus, scale (Planck 2018 and other parameter sets)

Z	redshift(s) to compute
<code>-objet, -object NAME</code>	object name: redshift requested from SIMBAD
<code>-modele, -model {planck18,planck15,wmap9,simple,perso}</code>	parameter set (default: planck18)
<code>-h0 H0</code>	H_0 in km/s/Mpc (custom model)
<code>-om OM</code>	total Ω_m (custom model)
<code>-ok OK</code>	Ω_k , curvature (planck18 or perso)
<code>-shoes</code>	add the SH0ES comparison (planck18)
<code>-sans-incertitudes, -no-uncertainties</code>	without uncertainties (faster)
<code>-csv FILE</code>	write the results to this CSV file
<code>-courbes, -curves ZMIN ZMAX N</code>	write the curves (z from ZMIN to ZMAX, N points) to the CSV file given by <code>-csv</code>
<code>-json</code>	JSON output (for scripts)

coupole sites

Observing sites (OpenStreetMap map), time zones, UTC and local time

<code>-ajouter, -add X</code>	add a site: ID NAME LAT LON ALT TIMEZONE [MPC]
<code>-supprimer, -delete ID</code>	delete an added site
<code>-heure, -time ISO_DATE</code>	show a UTC date (ISO) in site time and evening date
<code>-site ID</code>	site identifier (default: ohp)

coupole machine

Hardware diagnosis, chosen parallelism, graphics card, ASTAP

<code>-json</code>	JSON output (for scripts)
--------------------	---------------------------

CHAPTER 13

Evolution

Sources. Addresses (TAP, collection, portal, download URL rewriting, resolvers, collection point, repository, map tiles) are in `coupole/donnees/sources.json`. Three layers: shipped values; file published in the repository, fetched at startup when newer and passing the shape check (known keys, types, domains from an allow-list; no cryptographic signature); values forced by the user, never overwritten. Preferences > Sources: edit, back to default, test. If the image server moves, `ohp.telechargement.de/vers` rewrites the URLs.

Classification. Shipped table `data/classement.json` (194 checked names); published aliases; user corrections (*Correct the classification*, `coupole ohp classify`); name-shape rules; optional online resolution (JPL SBDB, CDS Sesame), cached; grouping by position on a known fixed object; otherwise « unclassified ». Every automatic classification is marked « to check ».

Database. Extra ObsCore columns ignored, missing ones replaced by a default, unknown instruments and filters stored apart, non-image data never flagged « daytime »; tested on simulated inventories.

CHAPTER 14

Incident reports and privacy

At first launch Coupole asks whether it may send anonymous reports: installation, sign of life (once a day), crash, native crash, frozen interface, written report. Content: version, system, processor, memory, graphics card, a random installation identifier, error trace with the home folder replaced by « ~ », the user name masked and deep paths shortened. Never a computer name, a user name or an image. Without consent nothing is sent: reports stay in `_rapports` in the settings folder. The choice can be changed (Help menu, `coupole reports -enable|-disable`).

CHAPTER 15

Writing a module

A module is a Python package with a `manifest.json` (identifier, version, French and English names, icon, texts, graphical and command-line entry points), discovered in `coupole/modules/`, through the `coupole.modules` entry point, or in the `modules/` folder of the settings. Texts through `tr()`, mandatory tooltips, long work off the GUI thread: see `CONTRIBUTING.md`.

CHAPTER 16

Troubleshooting

Symptom	What to do
Nothing starts (Windows) « Graphical interface unavailable »	Start menu > Coupole > Diagnostic: the error stays on screen. missing Qt libraries (Linux: <code>libxcb-cursor0</code> , <code>libegl1</code>); the command line still works.
Download fails	run again (resume); check the connection; Preferences > Sources > Test; server moved: URL rewriting.
« ASTAP not found »	ASTAP assistant; or <code>coupole astap -set PATH -catalog FOLDER</code> .
Incomplete catalogue	reinstall the catalogue (1,476 tiles expected).
Not enough space	choose another output folder; the space needed includes pending FITS.
Slow computer	economy mode, or fewer simultaneous conversions.
Unknown target name	Catalogue tab, « To check » box, <i>Correct the classification</i> .
Quality unavailable	<code>pip install sep</code> (included in the packages).

Sources

« OHP student observations » bank, Observatoire de Paris / PADC (Paris Astronomical Data Centre), public TAP service, Etalab 2.0 licence; official guide of the database (PADC-BDD-DU-ECU). XISF 1.0 specification (Pleiades Astrophoto). ASTAP: <https://www.hnsky.org/astap.htm>. Observatory codes: Minor Planet Center. SEP: Barbary (2016), Bertin & Arnouts (1996). Adaptive moments: Bernstein & Jarvis (2002). Map background: © OpenStreetMap contributors. SIMBAD and Sesame: CDS, Strasbourg (Wenger et al. 2000); Aladin Lite (CDS); NED (NASA/IPAC); JPL Small-Body Database (NASA/JPL). Cosmology: Planck Collaboration (2020, A&A 641, A6); Hinshaw et al. (2013, WMAP 9); Riess et al. (2022, SH0ES); Hogg (1999, astro-ph/9905116); Komatsu et al. (2011) for the massive-neutrino density; astropy.cosmology; « cosmologie-redshift » calculator (same author).